

Math 221 Homework 10

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Question 6.2. Let

$$T_N = \begin{pmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 2 & -1 \\ & & & -1 & 2 \end{pmatrix} \in \mathbb{R}^{N \times N}.$$

Prove the following triangular factorizations of T_N .

(1) *Cholesky factorization.* Show that $T_N = B_N^T B_N$ where B_N is upper bidiagonal with

$$B_N(i, i) = \sqrt{\frac{i+1}{i}}, \quad B_N(i, i+1) = -\sqrt{\frac{i}{i+1}}, \quad i = 1, \dots, N-1,$$

and $B_N(N, N) = \sqrt{\frac{N+1}{N}}$.

Proof. Since B_N is upper bidiagonal, $B_N^T B_N$ is tridiagonal. For $1 < i < N$,

$$(B_N^T B_N)_{ii} = B_N(i, i)^2 + B_N(i-1, i)^2 = \frac{i+1}{i} + \frac{i-1}{i} = 2,$$

and similarly $(B_N^T B_N)_{11} = B_N(1, 1)^2 = 2$ and $(B_N^T B_N)_{NN} = B_N(N, N)^2 + B_N(N-1, N)^2 = 2$. For $1 \leq i \leq N-1$,

$$(B_N^T B_N)_{i, i+1} = B_N(i, i)B_N(i, i+1) = \sqrt{\frac{i+1}{i}} \left(-\sqrt{\frac{i}{i+1}} \right) = -1,$$

and symmetry gives $(B_N^T B_N)_{i+1, i} = -1$. All other entries vanish, so $B_N^T B_N = T_N$. $\square$

(2) *LU factorization with partial pivoting.* Show that $T_N = L_N U_N$, where L_N, U_N are bidiagonal with

$$L_N(i, i) = 1, \quad L_N(i+1, i) = -\frac{i}{i+1}, \quad U_N(i, i) = \frac{i+1}{i}, \quad U_N(i, i+1) = -1$$

for $i = 1, \dots, N-1$, and $U_N(N, N) = \frac{N+1}{N}$.

Proof. Because L_N is lower bidiagonal and U_N is upper bidiagonal, $L_N U_N$ is tridiagonal. For $1 < i < N$,

$$(L_N U_N)_{ii} = L_N(i, i)U_N(i, i) + L_N(i, i-1)U_N(i-1, i) = \frac{i+1}{i} + \left(-\frac{i-1}{i}\right)(-1) = 2,$$

with the boundary cases $i = 1$ and $i = N$ handled similarly. For $1 \leq i \leq N-1$,

$$(L_N U_N)_{i, i+1} = L_N(i, i)U_N(i, i+1) = -1,$$

and

$$(L_N U_N)_{i+1, i} = L_N(i+1, i)U_N(i, i) = -\frac{i}{i+1} \cdot \frac{i+1}{i} = -1.$$

All other entries are zero, hence $L_N U_N = T_N$. □

(3) *Factorization* $T_N = D_N D_N^T$. Let $D_N \in \mathbb{R}^{N \times (N+1)}$ be upper bidiagonal with

$$D_N(i, i) = 1, \quad D_N(i, i+1) = -1, \quad i = 1, \dots, N.$$

Show that $T_N = D_N D_N^T$.

Proof. Each row i of D_N has only two nonzero entries, 1 in column i and -1 in column $i+1$. Thus

$$(D_N D_N^T)_{ii} = 1^2 + (-1)^2 = 2 \quad \text{for all } i.$$

For $1 \leq i \leq N-1$, the only common nonzero column of rows i and $i+1$ is column $i+1$, giving

$$(D_N D_N^T)_{i, i+1} = 1 \cdot 0 + (-1) \cdot 1 + 0 \cdot (-1) = -1,$$

and similarly $(D_N D_N^T)_{i+1, i} = -1$. If $|i - j| > 1$, rows i and j have disjoint supports, so $(D_N D_N^T)_{ij} = 0$. Hence $D_N D_N^T$ has diagonal entries 2, first sub- and superdiagonals equal to -1 , and zeros elsewhere, i.e. $D_N D_N^T = T_N$. □

Question 6.4.

1. Prove Lemma 6.2.
2. Prove Lemma 6.3.
3. Show that the Sylvester equation $AX - XB = C$ is equivalent to

$$(I_n \otimes A - B^T \otimes I_m) \text{vec}(X) = \text{vec}(C).$$

4. Show that

$$\text{vec}(AXB) = (B^T \otimes A) \text{vec}(X).$$

Proof. 1. (Lemma 6.2) Write $X = [x_1, \dots, x_n]$ with columns x_j . Then

$$\text{vec}(AX) = \begin{pmatrix} Ax_1 \\ \vdots \\ Ax_n \end{pmatrix} = \begin{pmatrix} A & & \\ & \ddots & \\ & & A \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = (I_n \otimes A) \text{vec}(X).$$

This is part (i). For part (ii), by definition of matrix multiplication, $(XB)_{ij} = \sum_{k=1}^n X_{ik} B_{kj}$. Arranging the entries in $\text{vec}(XB)$ shows that

$$\text{vec}(XB) = (B^T \otimes I_m) \text{vec}(X).$$

2. (Lemma 6.3) (i) The (i, j) -block of $A \otimes B$ is $A_{ij}B$, and similarly for $C \otimes D$. Thus the (i, j) -block of $(A \otimes B)(C \otimes D)$ is

$$\sum_k A_{ik} C_{kj} BD = (AC)_{ij} BD,$$

which is exactly the (i, j) -block of $(AC) \otimes (BD)$. Hence $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$.

(ii) If A and B are invertible, then using (i),

$$(A \otimes B)(A^{-1} \otimes B^{-1}) = (AA^{-1}) \otimes (BB^{-1}) = I_m \otimes I_n = I_{mn},$$

so $(A \otimes B)^{-1} = A^{-1} \otimes B^{-1}$.

(iii) View $A \otimes B$ as the block matrix whose (i, j) -block is $A_{ij}B$. Transposing gives a block matrix whose (i, j) -block is $(A_{ji}B)^T = A_{ji}B^T$, i.e. the (i, j) -block of $A^T \otimes B^T$. Hence $(A \otimes B)^T = A^T \otimes B^T$.

3. Apply vec to $AX - XB = C$:

$$\text{vec}(AX - XB) = \text{vec}(AX) - \text{vec}(XB) = (I \otimes A) \text{vec}(X) - (B^T \otimes I) \text{vec}(X) = \text{vec}(C),$$

using part 1. Thus

$$(I_n \otimes A - B^T \otimes I_m) \text{vec}(X) = \text{vec}(C).$$

4. Using part 1 and Lemma 6.3(i),

$$\text{vec}(AXB) = \text{vec}(A(XB)) = (I \otimes A) \text{vec}(XB) = (I \otimes A)(B^T \otimes I) \text{vec}(X) = (B^T \otimes A) \text{vec}(X).$$

□

Question 6.5. Suppose $A \in \mathbb{R}^{n \times n}$ is diagonalizable with

$$AX = X\Lambda_A, \quad X = [x_1, \dots, x_n], \quad \Lambda_A = \text{diag}(\alpha_i),$$

and $B \in \mathbb{R}^{m \times m}$ is diagonalizable with

$$BY = Y\Lambda_B, \quad Y = [y_1, \dots, y_m], \quad \Lambda_B = \text{diag}(\beta_j).$$

Prove:

1. The mn eigenvalues of $I_m \otimes A + B \otimes I_n$ are $\lambda_{ij} = \alpha_i + \beta_j$ with eigenvectors $z_{ij} = x_i \otimes y_j$. Equivalently,

$$(I_m \otimes A + B \otimes I_n)(Y \otimes X) = (Y \otimes X)(I_m \otimes \Lambda_A + \Lambda_B \otimes I_n).$$

2. The Sylvester equations

$$AX + XB^T = C, \quad AX + XB = C$$

are nonsingular (have a unique solution X for every C) iff $\alpha_i + \beta_j \neq 0$ for all eigenvalues α_i of A and β_j of B .

3. The mn eigenvalues of $A \otimes B$ are $\lambda_{ij} = \alpha_i \beta_j$ with eigenvectors $z_{ij} = x_i \otimes y_j$. Equivalently,

$$(B \otimes A)(Y \otimes X) = (Y \otimes X)(\Lambda_B \otimes \Lambda_A).$$

Proof. (1) Using the Kronecker product identities from Question 6.4,

$$(I_m \otimes A)(Y \otimes X) = Y \otimes AX = Y \otimes X\Lambda_A,$$

and

$$(B \otimes I_n)(Y \otimes X) = BY \otimes X = Y\Lambda_B \otimes X = (Y \otimes X)(\Lambda_B \otimes I_n).$$

Adding gives

$$(I_m \otimes A + B \otimes I_n)(Y \otimes X) = (Y \otimes X)(I_m \otimes \Lambda_A + \Lambda_B \otimes I_n).$$

Thus $Y \otimes X$ is a basis of eigenvectors, and the eigenvalues are the entries of $I_m \otimes \Lambda_A + \Lambda_B \otimes I_n$, namely $\alpha_i + \beta_j$ with eigenvectors $x_i \otimes y_j$.

- (2) From Question 6.4, vectorizing $AX + XB^T = C$ gives

$$(I_m \otimes A + B \otimes I_n) \text{vec}(X) = \text{vec}(C).$$

This linear system is nonsingular iff $\det(I_m \otimes A + B \otimes I_n) \neq 0$, i.e. iff 0 is not an eigenvalue of $I_m \otimes A + B \otimes I_n$. By part (1), its eigenvalues are $\alpha_i + \beta_j$, so the condition is exactly $\alpha_i + \beta_j \neq 0$ for all i, j .

For $AX + XB = C$ we similarly obtain

$$(I_m \otimes A + B^T \otimes I_n) \text{vec}(X) = \text{vec}(C),$$

whose eigenvalues are again $\alpha_i + \beta_j$ since B and B^T have the same eigenvalues. The same condition $\alpha_i + \beta_j \neq 0$ is therefore necessary and sufficient for nonsingularity.

(3) Using the mixed-product rule for Kronecker products,

$$(B \otimes A)(Y \otimes X) = BY \otimes AX = Y\Lambda_B \otimes X\Lambda_A = (Y \otimes X)(\Lambda_B \otimes \Lambda_A).$$

Hence $Y \otimes X$ is again a basis of eigenvectors, now for $B \otimes A$, with eigenvalues $\beta_j\alpha_i = \alpha_i\beta_j$ and eigenvectors $x_i \otimes y_j$. Since $A \otimes B$ and $B \otimes A$ are permutation-similar, they share the same multiset of eigenvalues, so the result holds for $A \otimes B$ as well. $\square$

Question 6.6 The line of matlab code is

$$x = x + (b - A * x) ./ \text{diag}(A);$$

and in practice, for the Poisson matrix, iterating this Jacobi step does converge at the rate predicted in section 6.5.4.

Question 6.11. Suppose R is upper triangular and nonsingular and C is upper Hessenberg. Show that RCR^{-1} is upper Hessenberg.

Proof. Since the inverse of an upper triangular matrix is again upper triangular, it is enough to show that left- and right-multiplication by an upper triangular matrix preserve the upper-Hessenberg structure.

Write $B = RC$. Then

$$B_{ij} = (RC)_{ij} = \sum_{k=1}^n r_{ik}c_{kj}.$$

If $i > j + 1$, split the sum into $k < i$ and $k \geq i$. For $k < i$ we have $r_{ik} = 0$ (upper triangular R), so those terms vanish. For $k \geq i > j + 1$ we have $k > j + 1$, hence $c_{kj} = 0$ (upper Hessenberg C). Thus $B_{ij} = 0$ whenever $i > j + 1$, so RC is upper Hessenberg.

Now let S be upper triangular and set $D = CS$. Then

$$D_{ij} = (CS)_{ij} = \sum_{k=1}^n c_{ik}s_{kj}.$$

If $i > j + 1$, only terms with $k \leq j$ can contribute, since $s_{kj} = 0$ for $k > j$. But for $k \leq j \leq i - 2$ we have $i > k + 1$, so $c_{ik} = 0$ by the Hessenberg property. Hence $D_{ij} = 0$ for all $i > j + 1$, so CS is also upper Hessenberg.

Applying this twice with $S = R$ and $S = R^{-1}$ gives that

$$RCR^{-1} = (RC)R^{-1}$$

is upper Hessenberg. □